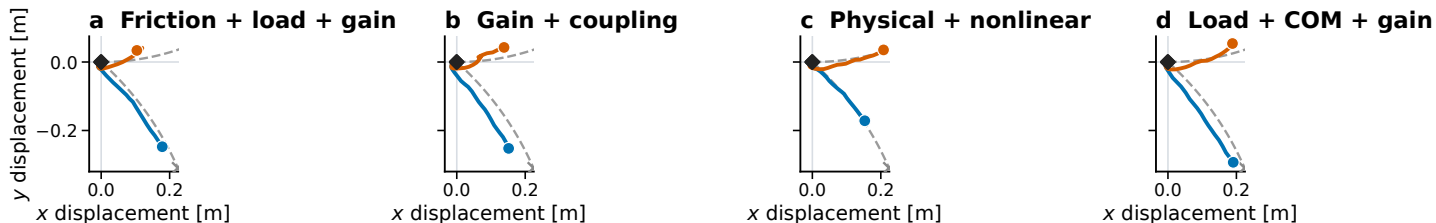


● Coupled command: [0.20, -0.18, -0.30]

● Forward-turn command: [0.35, 0.00, 0.50]

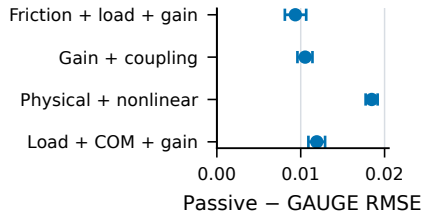
--- Ideal command response



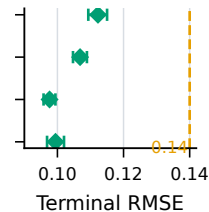
e Exact pre → post shifts

μ_s	0.92 → 0.28	0.84 → 0.84	0.80 → 0.42	0.72 → 0.72
μ_d	0.74 → 0.22	0.66 → 0.66	0.62 → 0.31	0.55 → 0.55
Payload [kg]	0.0 → 2.8	0.0 → 0.0	0.0 → 2.2	0.0 → 3.0
COM [mm]	0 → 28	0 → 0	0 → -22	0 → -32
Gain (pre)	1.05--1.15	1.05--1.15	1.05--1.15	1.05--1.15
Gain (post)	.65--.72	.65--.72	.65--.72	.65--.72
$ c_{ij} _{\max}$ (pre)	≤ .04	≤ .04	≤ .04	≤ .04
$ c_{ij} _{\max}$ (post)	≤ .12	≤ .12	≤ .12	≤ .12
	Friction, load, gain	Gain, coupling	Physical, nonlinear	Load, COM, gain

f Early recovery



g Absolute gate



h Recovery-selector ablation (four contexts averaged within paired seed)

